

0-getting-started-tutorial-how-to

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1 Chapter 0. Getting Started-Tutorial How-To

You need to run this initial notebook first to ensure that the necessary requirements and environment variables are set up for all notebooks in this cookbook. In this section, we will introduce Solar, a large language model (LLM) developed by Upstage. This notebook covers the basics of setting up Solar, understanding its core components, and running a simple “Hello, World!” example to demonstrate its basic functionality.

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-

1.2 Setting

1.2.1 `UPSTAGE_API_KEY`

To obtain your Upstage API key, follow these steps:

1. Visit the Upstage AI console at <https://console.upstage.ai>.
2. Sign up for an account if you don't already have one.
3. Log in to your account.
4. Navigate to the API Key section.
5. Generate your API Key.
6. Copy the key and save it securely.

1.2.2 How to run Solar Pro

- Install the required dependencies by running the following command(**Shift(or Cmd) + Enter**):

```
[ ]: %%capture
!pip install openai pickleshare langchain-upstage llama_index.llms.upstage_
↳ llama_index
```

```
[ ]: import pickleshare
import os
from dotenv import load_dotenv
load_dotenv()

UPSTAGE_API_KEY = os.getenv("UPSTAGE_API_KEY")

# To load the UPSTAGE_API_KEY as a variable across all notebooks in this_
↳ cookbook.
%store UPSTAGE_API_KEY
```

Stored 'UPSTAGE_API_KEY' (str)

```
[ ]: # Python
from openai import OpenAI

client = OpenAI(
    api_key = UPSTAGE_API_KEY,
    base_url = "https://api.upstage.ai/v1/solar",
)

response = client.chat.completions.create(
    model= 'solar-pro',
    messages= [
        {
            "role": "user",
            "content": "Hi, how are you?"
        }
    ]
)

print(response.choices[0].message.content, '\n\n')
```

Hello! I'm just a computer program, so I don't have feelings, but I'm functioning perfectly. How can I assist you today?

1.3 Alternative Options for the Task

1.3.1 Request the API with the Example Code

```
[ ]: # LangChain
from langchain_upstage import ChatUpstage

client = ChatUpstage(api_key=UPSTAGE_API_KEY, model="solar-pro")

response = client.invoke("Hi, how are you?")
print(response.content, '\n\n')
```

Hello! I'm just a computer program, so I don't have feelings, but I'm functioning perfectly and ready to assist you. How can I help you today?

```
[ ]: # LlamaIndex
from llama_index.llms.upstage import Upstage
from llama_index.core.llms import ChatMessage

client = Upstage(api_key=UPSTAGE_API_KEY, model="solar-pro")

message = []
message.append(ChatMessage(role="user", content="Hi, how are you?"))

response = client.chat(messages=message)
print(response.message.content, '\n\n')
```

Hello! I'm just a computer program, so I don't have feelings, but I'm functioning perfectly. How can I assist you today?

Upstage LangChain integrations

API	Description	Import	Example Usage
Chat	Build assistants using Solar Mini Chat	from langchain_upstage import ChatUpstage	Go
Text Embedding	Embed strings to vectors	from langchain_upstage import UpstageEmbeddings	Go
Groundedness Check	Verify groundedness of assistant's response	from langchain_upstage import UpstageGroundednessCheck	Go

API	Description	Import	Example Usage
Layout Analysis	Serialize documents with tables and figures	from langchain_upstage import UpstageLayoutAnalysisLoader	Go

See [documentations](#) for more details about the features.

```
[ ]: # Quick hello Langchain Upstage
# Make sure you have the langchain-upstage package installed
# Make sure UPSTAGE_API_KEY is set in the environment variables
import os
from langchain_upstage import ChatUpstage

os.environ["UPSTAGE_API_KEY"] = UPSTAGE_API_KEY

llm = ChatUpstage(model="solar-1-mini-chat")
print(llm.invoke("Hello Mini!"), '\n\n')
```

```
content='Hello! How can I assist you today?' additional_kwargs={'refusal': None}
response_metadata={'token_usage': {'completion_tokens': 10, 'prompt_tokens': 13,
'total_tokens': 23, 'completion_tokens_details': None, 'prompt_tokens_details':
None}, 'model_name': 'solar-mini-240612', 'system_fingerprint': None,
'finish_reason': 'stop', 'logprobs': None}
id='run-a64bfe7c-65c5-4188-a3f0-3b13778906cb-0' usage_metadata={'input_tokens':
13, 'output_tokens': 10, 'total_tokens': 23, 'input_token_details': {},
'output_token_details': {}}
```

```
[ ]: from langchain_upstage import ChatUpstage

llm = ChatUpstage(model="solar-1-mini-chat-ja")
print(llm.invoke("      "), '\n\n')
```

```
content='      ' additional_kwargs={'refusal': None}
response_metadata={'token_usage': {'completion_tokens': 10, 'prompt_tokens': 15,
'total_tokens': 25, 'completion_tokens_details': None, 'prompt_tokens_details':
None}, 'model_name': 'solar-mini-ja-240612', 'system_fingerprint': None,
'finish_reason': 'stop', 'logprobs': None}
id='run-84880253-e21e-4e60-86c6-3ec4d2de4e8e-0' usage_metadata={'input_tokens':
15, 'output_tokens': 10, 'total_tokens': 25, 'input_token_details': {},
'output_token_details': {}}
```

```
[ ]: # Solar Groundedness Check
from langchain_upstage import UpstageGroundednessCheck

groundedness_check = UpstageGroundednessCheck()

context = "Solar is a language model that can generate human-like text. It is_
↳developed by Upstage."
print(groundedness_check.invoke(
    {"context": context, "answer": "Solar is created by Upstage."}
), '\n\n')
```

grounded

```
[ ]: print(groundedness_check.invoke({"context": context, "answer": "Solar is_
↳created by OpenAI."})), '\n\n')
```

notGrounded

```
[ ]: from langchain_upstage import UpstageEmbeddings

embeddings_model = UpstageEmbeddings(model="solar-embedding-1-large")
embeddings_model.embed_documents(
    [
        "What is the best season to visit Korea?",
    ]
)
```

```
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Next: Chapter 1. Basic Prompt Structure